

# Hoa Huynh

Email: [huynhthaihoa1995@gmail.com](mailto:huynhthaihoa1995@gmail.com)

LinkedIn: [linkedin.com/in/huynhthaihoa](https://www.linkedin.com/in/huynhthaihoa)

GitHub: [github.com/huynhthaihoa](https://github.com/huynhthaihoa)

Google Scholar: [scholar.google.com/citations?user=abuFJTkAAAAJ&hl=en](https://scholar.google.com/citations?user=abuFJTkAAAAJ&hl=en)

Seoul, South Korea

Phone: +82-10-2907-2895

Portfolio: [huynhthaihoa.github.io](https://huynhthaihoa.github.io)

LeetCode: [leetcode.com/u/huynhthaihoa](https://leetcode.com/u/huynhthaihoa)

## ABOUT

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Computer Vision Engineer with more than seven years of experience developing and deploying real-time deep learning systems on edge devices for automotive safety and surveillance. Experienced in 2D/3D perception, inference optimization, multi-sensor integration, and edge AI deployment.

## RESEARCH INTERESTS

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Visual SLAM, 3D Vision, Robotics, Automotive, Edge AI Deployment

## SKILLS

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**Programming Languages:** C, C++, Python, C#, MATLAB

**Perception & Deep Learning:** Open3D, PCL, OpenCV, PyTorch, TensorFlow, Darknet

**Edge AI Deployment:** TI Edge AI/TIOVX (SK-TDA4VM, TDA4AEN), NVIDIA Jetson Nano/Xavier NX (TensorRT, DeepStream), Samsung Exynos V920, ONNX Runtime, TFLite

**Sensors & Calibration:** RGB/IR pinhole/fisheye, FLIR thermal, Intel RealSense, LUCID Helios2 ToF, Ouster LiDAR, ROS

**MLOps & Tooling:** Git, SVN, Docker, CMake, TensorBoard, DVC, W&B, MLflow, Jira, Confluence

**Databases:** Microsoft SQL Server, MySQL, SQLite, PostgreSQL, MongoDB

**Additional Languages & Frameworks:** Node.js, React, JavaScript, Java, LaTeX, WPF, Qt, FFmpeg

## EXPERIENCE

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### DeltaX

Seoul, South Korea

*Development Team Lead & AI Researcher*

*June 2022 – July 2026*

- Led an R&D team from December 2023, driving the architecture, technical roadmap, and end-to-end delivery of an In-Cabin Monitoring System (ICMS) that aligned with Euro NCAP standards, while coordinating module development, milestone tracking, and client demonstrations across cross-functional teams.
- Architected a real-time edge AI pipeline using a single RGB/IR fisheye camera, C/C++ inference on an SoC (System-on-Chip), and a Node.js/React web visualization UI.
- Integrated 8 deep learning models (object detection, 2D/3D body keypoint estimation, facial landmark detection, gaze estimation, and image classification), achieving an average of 14 FPS on an 8-TOPS edge platform and 9 FPS on a 4-TOPS edge platform.
- Delivered 4 Proof-of-Concept (PoC) projects for 2 Korean automotive vendors within 3–9-month timelines: 42dot (Phase 1: January – June 2024; Phase 2: August – October 2024), Pyeong Hwa Automotive (Phase 1: May 2024 – January 2025; Phase 2: July – October 2025).
- Customized PyTorch models and optimized Python/C++ integration pipelines for SoC targets, deploying on TI SK-TDA4VM/TDA4AEN using Edge AI + TIOVX frameworks, and on Samsung Exynos V920 using an ONNX Runtime-based framework.
- Developed the 3D perception stack for worker localization, 3D display, and ADAS applications (Advanced Driver-Assistance Systems): monocular depth estimation (with pseudo-LiDAR conversion) trained under LiDAR/RGB-D supervision, semantic segmentation, visual SLAM, and real-time multi-camera stitching from fisheye and pinhole inputs.
- Designed a slicing-based preprocessing scheme derived from SAHI (“Slicing Aided Hyper Inference and Fine-tuning for Small Object Detection”) to raise recall on small objects for YOLOv8-p2/YOLOv9n models while maintaining real-time inference latency.
- Built the team’s multi-sensor calibration, data collection, and analysis tooling (ROS, MATLAB, Python), and assembled real-world ADAS/in-cabin datasets across diverse subjects and lighting conditions.

## Smartdeep

Research Engineer

Seoul, South Korea

May 2021 – May 2022

- Built a 16-channel CCTV surveillance system in C++ (OpenCV, FFmpeg) with a C# WPF front end, aligned with KISA Intelligent CCTV certification requirements for moving- and abnormal-object detection, recording, and event search.
- Owned dataset curation, training, and validation pipelines for Darknet YOLOv4 detection models.

## Ubay Solution

AI Research Intern

Seoul, South Korea

December 2020 – April 2021

- Developed an optical character recognition (OCR) system for deployment at Busan Port, enabling automated detection and recognition of truck license plate numbers and container codes.
- Benchmarked object detection models (SSD, YOLOv4, Faster R-CNN, etc.) on a custom dataset.
- Applied data augmentation and web crawling techniques to increase dataset size and quality.
- Implemented a DeepStream + TensorRT pipeline on NVIDIA Jetson Nano and Jetson Xavier NX for model deployment and real-time inference.

## ANDA Lab – Soongsil University

Research Assistant

Seoul, South Korea

September 2018 – August 2020

- Conducted research and experimental analysis using Python, OpenCV, and MATLAB to diagnose issues in a vehicle localization system centered on taillight detection and processing.
- Co-authored papers proposing solutions to address the identified issues.

## Global Cybersoft JSC

Software Engineer Trainee

Ho Chi Minh City, Vietnam

March 2017 – August 2017

- Developed C++ Microsoft Foundation Classes (MFC) plugins for SCREEN Holdings Co.'s industrial printer system (Japan).
- Developed Adobe Illustrator plugins using the Adobe Illustrator SDK, C++, and JavaScript for an advertising agency (Japan).
- Improved code quality and team collaboration by participating in Jira-based peer code reviews.
- Gained hands-on experience in an Agile software development environment.

## EDUCATION

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### Soongsil University

Master of Engineering, Information and Communication Technology

Seoul, South Korea

September 2018 – August 2020

- **GPA:** 4.01/4.5
- **Thesis:** LED Taillight Matching Algorithm for Nighttime Vehicle Positioning
- **Ref:** [oasis.ssu.ac.kr/search/i-discovery/edsker:edsker.000004851619?type=eds](https://oasis.ssu.ac.kr/search/i-discovery/edsker:edsker.000004851619?type=eds) (English)
- **Supervisor:** Professor Myungsik Yoo

### University of Information Technology – Vietnam National University

Bachelor of Engineering, Software Engineering

Ho Chi Minh City, Vietnam

August 2013 – June 2018

- **GPA:** 8.6/10 (Salutatorian Honors)
- **Thesis:** Blood Vessel Segmentation on DICOM Images
- **Ref:** [drive.google.com/file/d/1RylFqRy-xywZG1zEWEX2l3B49T7XbFvf/view](https://drive.google.com/file/d/1RylFqRy-xywZG1zEWEX2l3B49T7XbFvf/view) (Vietnamese)
- **Supervisor:** Professor Thi-Vuong Pham

## PUBLICATIONS

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(*J*: International Journal, *W*: International Workshop, *K*: Korean Journal/Conference)

**K.** Uddin S M Nadim et al. (including **Huynh Thai Hoa**), “A Railway Framework for Person (Worker) Localization based on Depth Estimation and GPS”, *The Korean Institute of Defense Technology Fall Conference* (2024). Ref: [kidet.or.kr/index.php?page=view&pg=&idx=1259&hCode=BOARD&bo\\_idx=1&sfl=&stx=](https://kidet.or.kr/index.php?page=view&pg=&idx=1259&hCode=BOARD&bo_idx=1&sfl=&stx=)

**W.** Jamie Spencer et al. (including **Huynh Thai Hoa**), “The Second Monocular Depth Estimation Challenge”, *Conference on Computer Vision and Pattern Recognition Workshops* (2023). Ref: [arxiv.org/abs/2304.07051](https://arxiv.org/abs/2304.07051)

**J. Thai-Hoa Huynh**, Myungsik Yoo, “A Taillight Matching and Pairing Algorithm for Stereo-Vision-Based Night-time Vehicle-to-Vehicle Positioning”, *MDPI Applied Sciences* (2020). Ref: [mdpi.com/2076-3417/10/19/6800](https://mdpi.com/2076-3417/10/19/6800)

**K. Thai-Hoa Huynh**, Myungsik Yoo, “Nighttime LED Taillight Detection Method with Optical Camera Communication System”, *The Journal of Korean Institute of Communications and Information Sciences* (2020). Ref: [dbpia.co.kr/journal/articleDetail?nodeId=NODE10440000](https://dbpia.co.kr/journal/articleDetail?nodeId=NODE10440000)

**J. Thai-Hoa Huynh**, Tuan-Anh Pham, Myungsik Yoo, “Detection Algorithm for Overlapping LEDs in Vehicular Visible Light Communication System”, *IEEE Access* (2019). Ref: [ieeexplore.ieee.org/document/8792184](https://ieeexplore.ieee.org/document/8792184)

## SELECTED PROJECTS

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**State Estimation Simulations:** from-scratch Lie-group state-estimation simulations – EKF, Invariant EKF, UKF, factor-graph and incremental (iSAM) optimization, bundle adjustment. [GitHub](#)

**Monocular Metric Depth Estimation:** PyTorch pipeline estimating per-pixel metric depth from a single image, with pluggable encoders (EfficientNet, ConvNeXt, DINOv2, YOLO-based) and decoders (NeWCRFs, DPT, PixelFormer), trained on NYUv2/KITTI dataset with quantization-aware training support and deployed via ONNX/TFLite export. [GitHub](#)

**Notch-Based Point Cloud Registration:** C++ point-cloud registration aligning 3D scans to a reference model via notch-keypoint correspondence matching, for industrial inspection and quality verification. [GitHub](#)

**LiDAR-Based Ego Lane Detection:** classical signal-processing pipeline (ground extraction, intensity analysis, polynomial fitting) for interpretable ego-lane detection directly from LiDAR point clouds. [GitHub](#)

**Vision Encoder Playground:** curated collection of pre-trained vision encoder (backbones) from detection and classification models, exposed as reusable feature extractors for downstream computer vision tasks. [GitHub](#)

(Full project list available on portfolio: [huynhthaihoa.github.io/#projects](https://huynhthaihoa.github.io/#projects))

## SELECTED CERTIFICATIONS

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### Hugging Face:

- Fine-tuning Language Models (2025)
- AI Agents Fundamentals (2025)
- Fundamentals of Group Relative Policy Optimization (2025)

### Coursera:

- Introduction to Machine Learning in Production (2024)

### HackerRank:

- Problem Solving (Intermediate) (2024)

### LeetCode:

- 500 Days Badge (2026)
- Introduction to Pandas (2024)

### Kaggle:

- Computer Vision (2022)
- Advanced SQL (2022)
- Intermediate Machine Learning (2020)

(Full certification list available on portfolio: [huynhthaihoa.github.io/#certifications](https://huynhthaihoa.github.io/#certifications))